

Radical Ad Tech

The Counterintuitive Guide to Ad Performance

Who am I?

Senior Software Engineer, Ad Tech Specialist

What I've learned testing ad tech best practices:

- Cut infrastructure costs by hundreds of thousands
- Served billions of impressions faster
- Proved the "performance vs. revenue" trade-off is BS

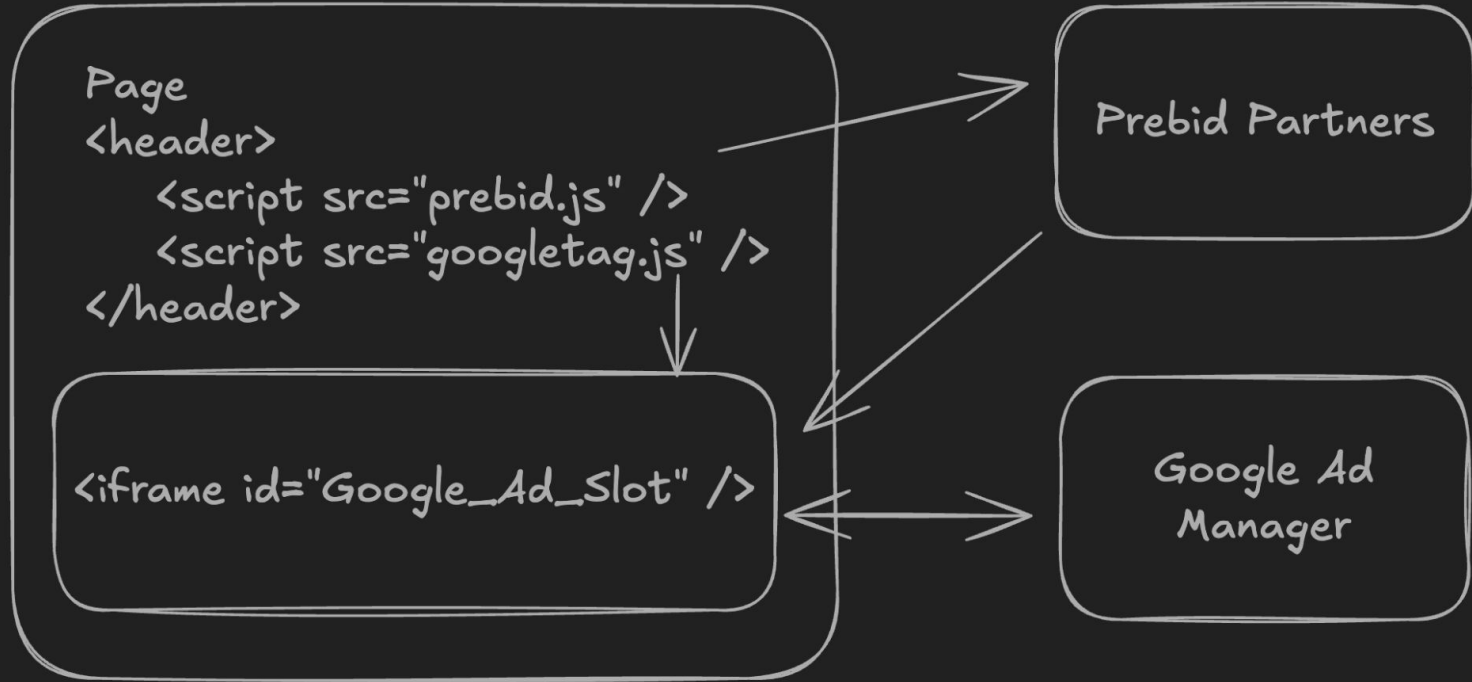


On Each Page With Ads There is an Auction

- Advertisers submit bids for ad space with target audience criteria
- Platform runs instant auction when user loads page
- Winner determined by bid amount & ad quality score
- Whole process happens in milliseconds



The Anatomy of an Ad Request



Ad Slots

- The element on your page google fills in with ads
- You can see them on most pages with `googletag.openConsole()`

18 Dynamic Allocation
18190176 > AdThrive_Content_6 >
541917f5a90318f9194874cf
AdThrive_Content_6_desktop
Creative: 138506235943
Line Item: 4423950873
Click for more data in Delivery Tools

Robust charting with 400+ technical studies on thinkorswim® desktop.

Up-to-the-minute built-in news and insights.

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4 Direct Sold
18190176 > AdThrive_Below_Post_1 > 541917f5a90318f9194874cf
AdThrive_Below_Post_1_desktop
Creative: 138496895583
Line Item: 6051712294
Click for more data in Delivery Tools

THE WIZARDING WORLD
OF HARRY POTTER™

UNIVERSAL

Ad Slots (25) Page Request Configuration

AUDIT THIS PAGE OPEN IN DELIVERY TOOLS

Inspect Creative

2 18190176 > AdThrive_Footer_1 >
541917f5a90318f9194874cf
Div: AdThrive_Footer_1_desktop Slot size: 728x90, 320x50, 970x90, 300x50, 320x100, 468x60, 1x1, fluid Out of Page: False Service: Ad Manager

Overlay status: **Displayed**
3035 ms to fetch
Ad fetch count: 1
Iframe type: **SafeFrame**
Line Item ID: 4423950873 Creative ID: 138506235943
Query ID: COjDnoK0uZIDFQTDrQAdxs8gXQ

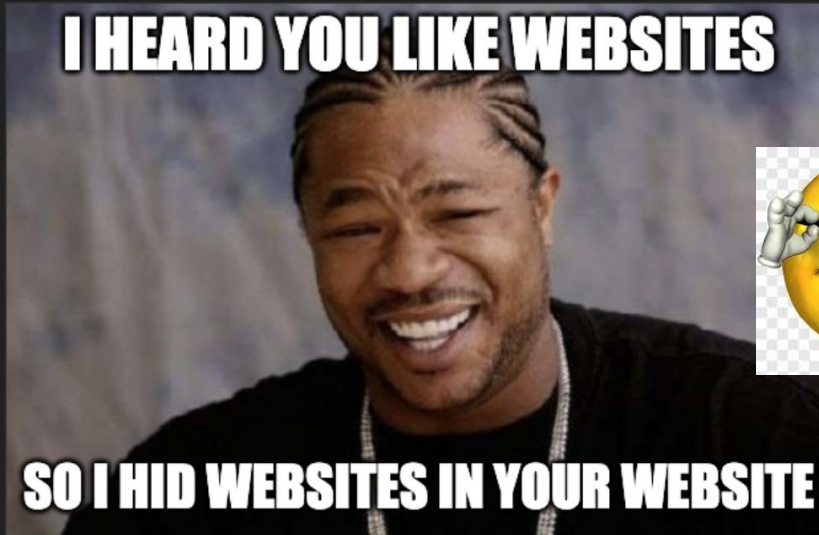
The above Query ID can be sent to our support teams if you have concerns about the content or behavior of the delivered creative.

[Open Targeting Info +](#)

[Open Ad Unit in Ad Manager](#) | [Delivery Diagnostics](#) | [Open Creative in New Window](#) | [Inspect Creative](#)

iframes!

- iframes are good for isolation
- Ads also typically add many hidden iframes

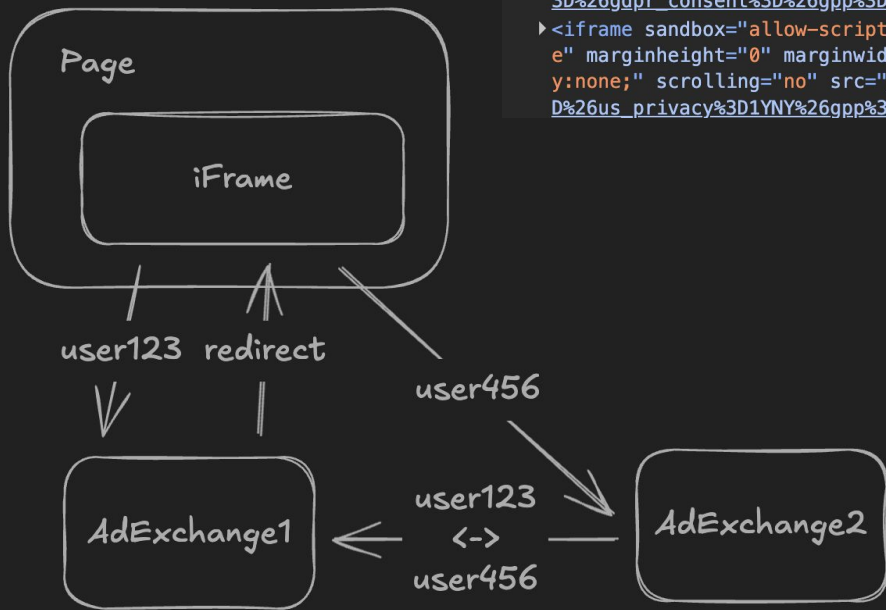


```

<iframe src="https://s.amazon-adsystem.com/iu3?cm.
style="display: none;">...</iframe> (ad)
<iframe name="google_ads_top_frame" id="google_ads
px; top: -999px; width: 0px; height: 0px;">...</if
<iframe src="https://www.google.com/recaptcha/api
</iframe> (ad)
<iframe height="0" width="0" frameborder="0" src="
3A%2F%2Fwww.Mjiizgtbchh0W3tx4qkYgU1a-hln3cUIgJXC64
...</iframe> (ad)
<script type="text/javascript" src="https://pghub.
<div id="AdThrive_Footer_1_desktop" class="adthrive
ve-footer-desktop adthrive-sticky" data-google-que
<div class="adthrive-sticky-outstream adthrive-st
m-idle" style="min-height: auto;">...</div>
</body>
<iframe id="goog_plcm_frame" src="https://cm.g.doub
C7&url=https%3A%2F%2Fwww.spendwithpennies.com%2Fbee
<iframe name="ifrm_pubmatic" src="https://ads.pubma
style="display: none;">...</iframe> (ad)
<iframe sandbox="allow-scripts allow-same-origin" i
marginheight="0" marginwidth="0" width="0" hspace="
one;" scrolling="no" src="https://eb2.3lift.com/syn
privacy%3D1YNY%26gpp%3D%26gpp_sid%3D%26f%3Db%26uid%
<iframe sandbox="allow-scripts allow-same-origin" i
marginheight="0" marginwidth="0" width="0" hspace="
one;" scrolling="no" src="https://u.openx.net/w/1.0
t%3D%26gpp%3D%26gpp_sid%3D%26f%3Db%26uid%3D%7BOPENX
<iframe sandbox="allow-scripts allow-same-origin" i
marginheight="0" marginwidth="0" width="0" hspace="
one;" scrolling="no" src="https://ssum.casalemedia.
3D%26gdp_r consent%3D%26gpp%3D%26gpp_sid%3D%26f%3Db%
<iframe sandbox="allow-scripts allow-same-origin" i
e" marginheight="0" marginwidth="0" width="0" hspace
y:none;" scrolling="no" src="https://u.openx.net/w/
D%26us_privacy%3D1YNY%26gpp%3D%26gpp_sid%3D%26f%3Db

```


Usersync

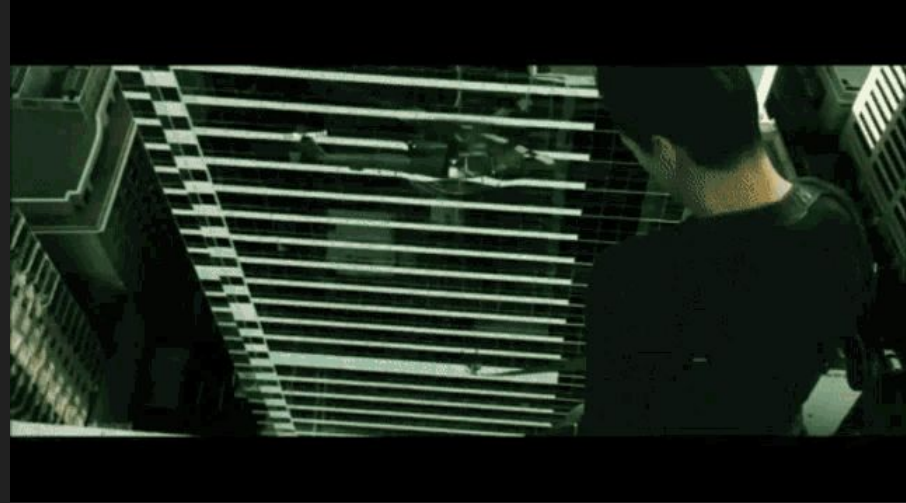


```
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marginheight="0" marginwidth="0" width="0" hspace="0" vspace="0" height="0" style="height:0px;width:0px;display:n
one;" scrolling="no" src="https://eb2.3lift.com/sync?gdpr=&cmp_cs=&us_privacy=1YNY&gpp=&gpp_sid=&...onsent%3D%26us
privacy%3D1YNY%26gpp%3D%26gpp_sid%3D%26f%3Db%26uid%3D%24UID">...</iframe> (ad)
><iframe sandbox="allow-scripts allow-same-origin" id="83813d21c234146e" frameborder="0" allowtransparency="true"
marginheight="0" marginwidth="0" width="0" hspace="0" vspace="0" height="0" style="height:0px;width:0px;display:n
one;" scrolling="no" src="https://u.openx.net/w/1.0/cm?id=891039ac-a916-42bb-a651-4be9e3b201da&...%3D%26gdpr_consen
t%3D%26gpp%3D%26gpp_sid%3D%26f%3Db%26uid%3D%7B0PENX_ID%7D">...</iframe> (ad)
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marginheight="0" marginwidth="0" width="0" hspace="0" vspace="0" height="0" style="height:0px;width:0px;display:n
one;" scrolling="no" src="https://ssum.casalemedia.com/usermatch?s=205820&gdpr=&gdpr_consent=&us_...er%3Dix%26gdpr%
3D%26gdpr_consent%3D%26gpp%3D%26gpp_sid%3D%26f%3Db%26uid%3D">...</iframe> (ad)
><iframe sandbox="allow-scripts allow-same-origin" id="1339a0fed992bd9c28" frameborder="0" allowtransparency="tru
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y:none;" scrolling="no" src="https://u.openx.net/w/1.0/cm?id=891039ac-a916-42bb-a651-4be9e3b201da&...gdpr_consent%3
D%26us_privacy%3D1YNY%26gpp%3D%26gpp_sid%3D%26f%3Db%26uid%3D">...</iframe> (ad)
```

- Your browser loads the iframe from the ad exchange domain
- The ad exchange reads **its own** cookie from your browser
- The iframe **redirects** to the partner's domain
- The partner reads **their own** cookie and creates a mapping
- **Both companies** now know these IDs represent the **same user**

Destroy your Invisible iFrames

- Hidden frames, like usersync frames, **usually aren't useful after page load completes**
- Frames can be removed from the DOM manually using the signatures of hidden frames
- In testing resulted in a **memory reduction of 40%**
- (With no drop in ad revenue)





- Measures how much your visible content changes position as it loads
- One of the Google **Core Web Vitals**
- So it **affects your search rankings!**

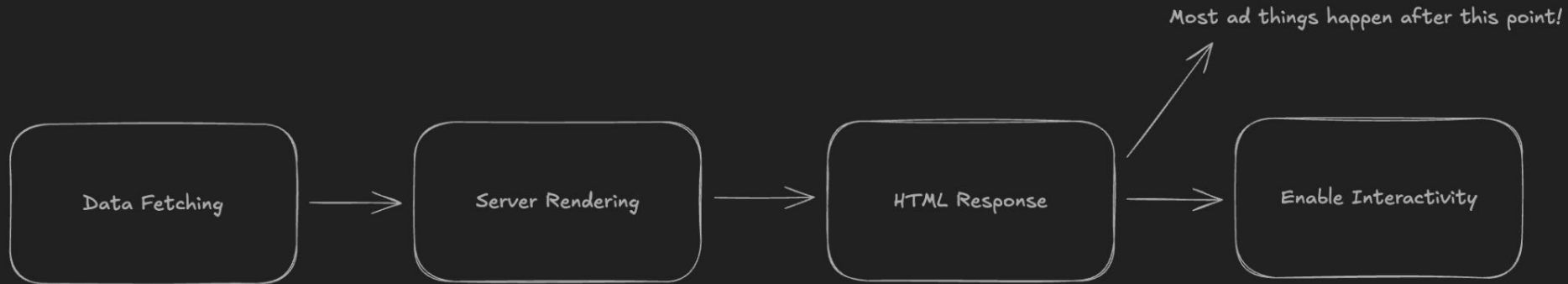


Eliminate Cumulative Layout Shift

- Collapsible ad slots are the **biggest culprit** of this
- Don't use them!
- Render the space **optimistically**
- If you end up with no fill, use a backstop "house" ad
- Upsell your own subscriptions

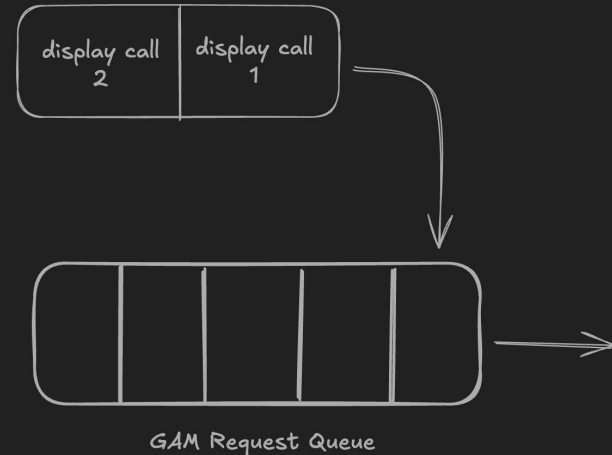
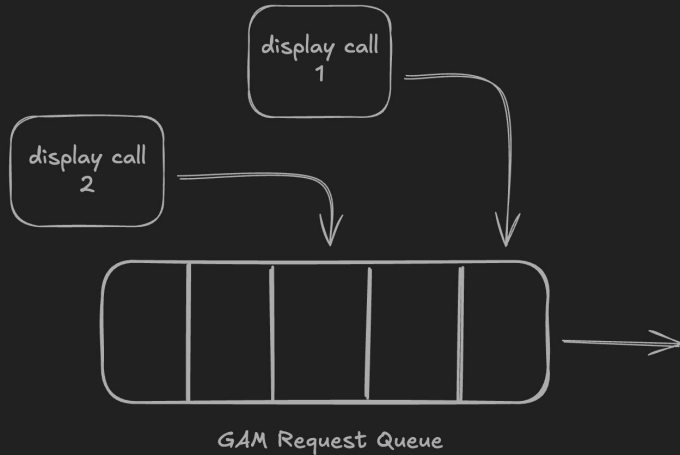
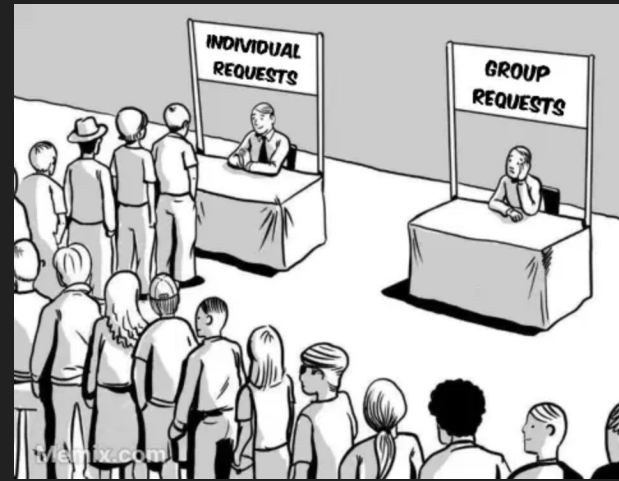
Working with Server-Side Rendering

- Loads dynamic content immediately with the page response
- Good for UX and Search Engine Optimization
- Ad scripts are **heavy** and **secondary** to the latter concerns
- They also rely on client side context (e.g. cookies) to run properly
- Ad scripts can **start loading after HTML response**
- Slots can't be displayed until **after interactivity is enabled** (stable DOM)

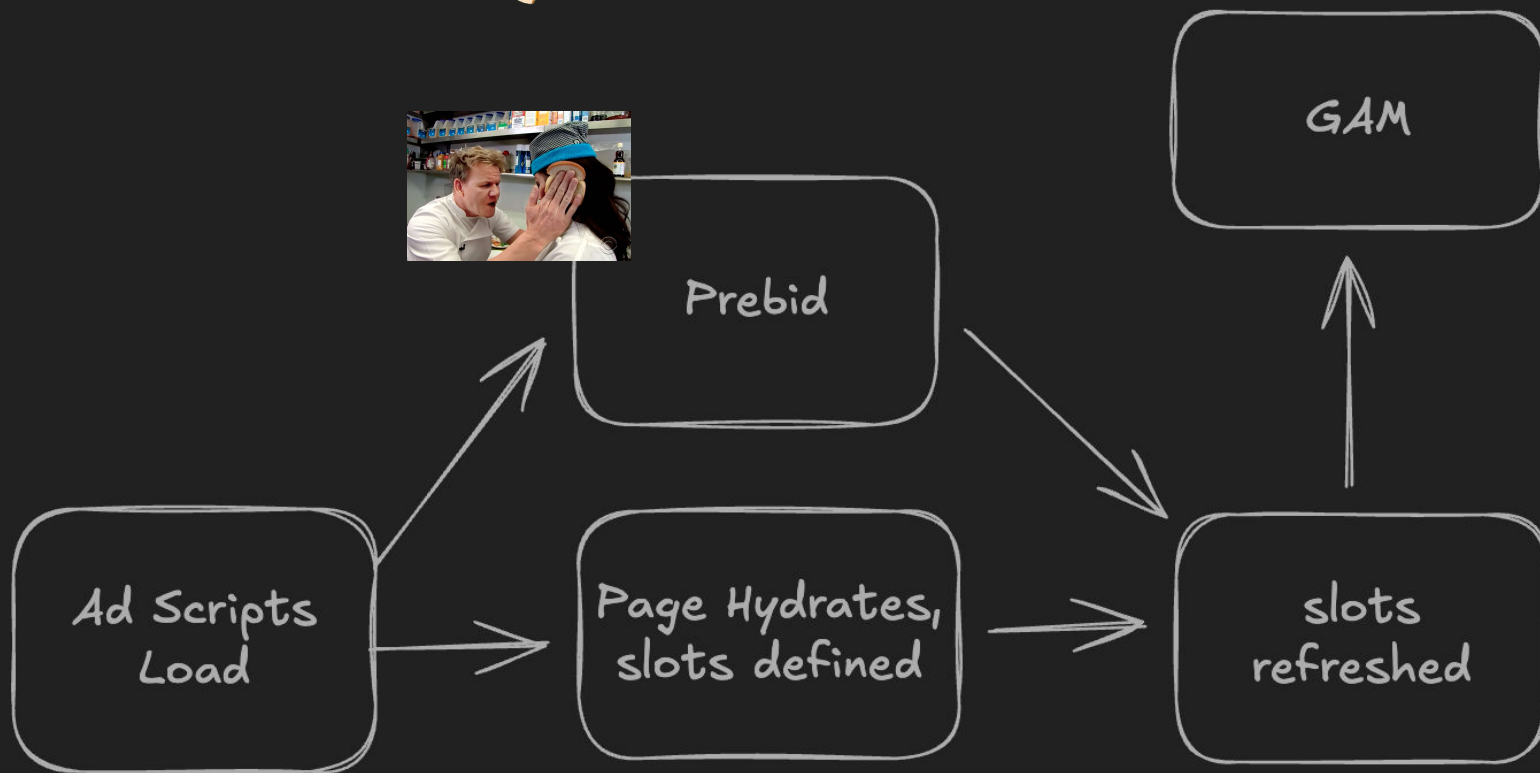


Single Request Architecture

- Adds GAM requests to a **client side queue**
- Fires **one request** as soon as everything's ready
- Queues them **in a block** on GAM's server

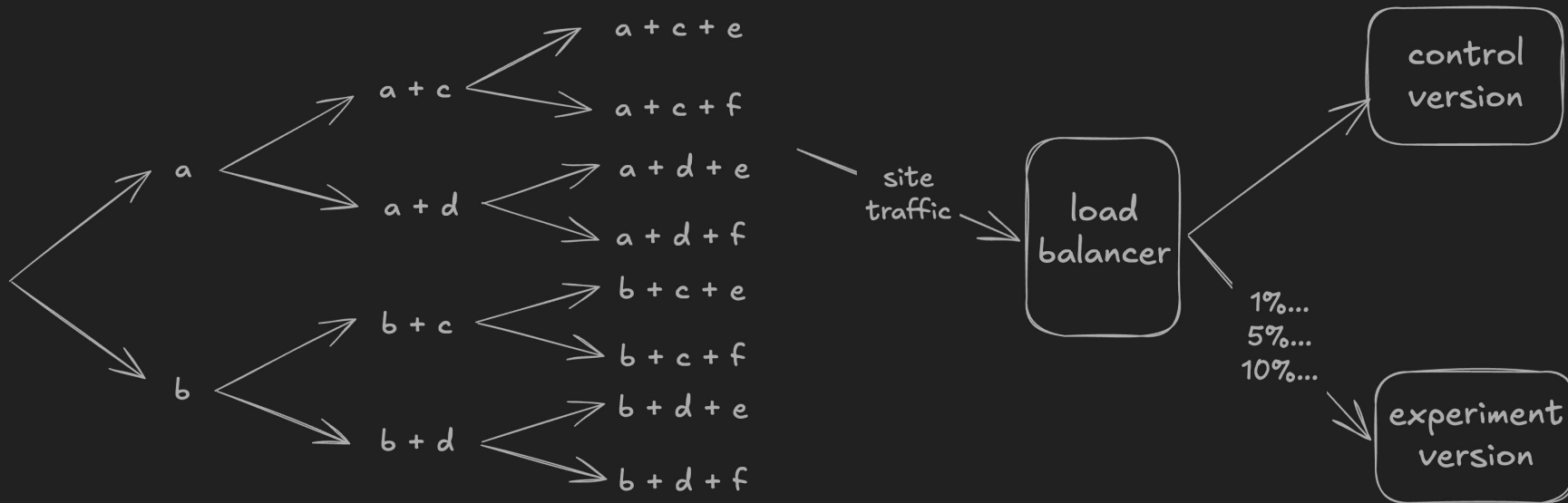


HB + SSR + SRA = 🤞



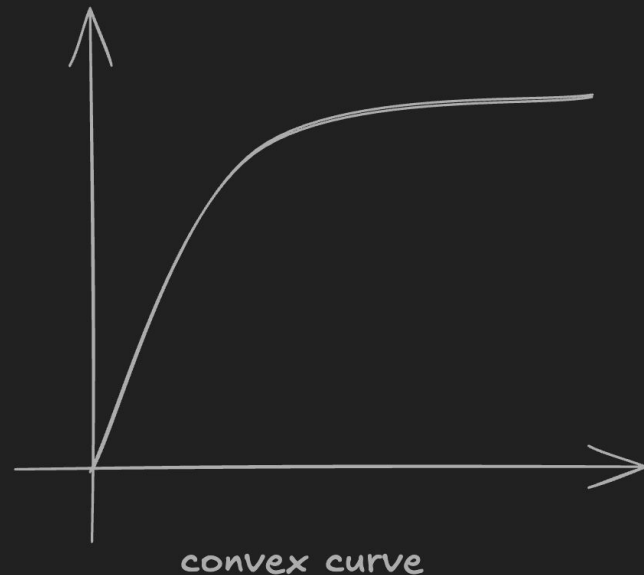
A/B Testing

- A lot of ad tech is about working with black-box third party tools.
- Focus on **desired outcomes**
- Scale your A/B testing with **canary deploys**



Are you *sure* it works? Really sure?

- “The absence of evidence is not the evidence of absence”
- *Donald Rumsfeld*
- You can't calculate the probability of what you haven't imagined or observed, but you **can** measure how your **system's payoff changes** as a **function of volatility** or extreme events, regardless of what those events are.
- **Use a “stop loss”**: bound your losses vs. unbounded upside.
- **Build in redundancy**: having many bidders and bid sources can help with this

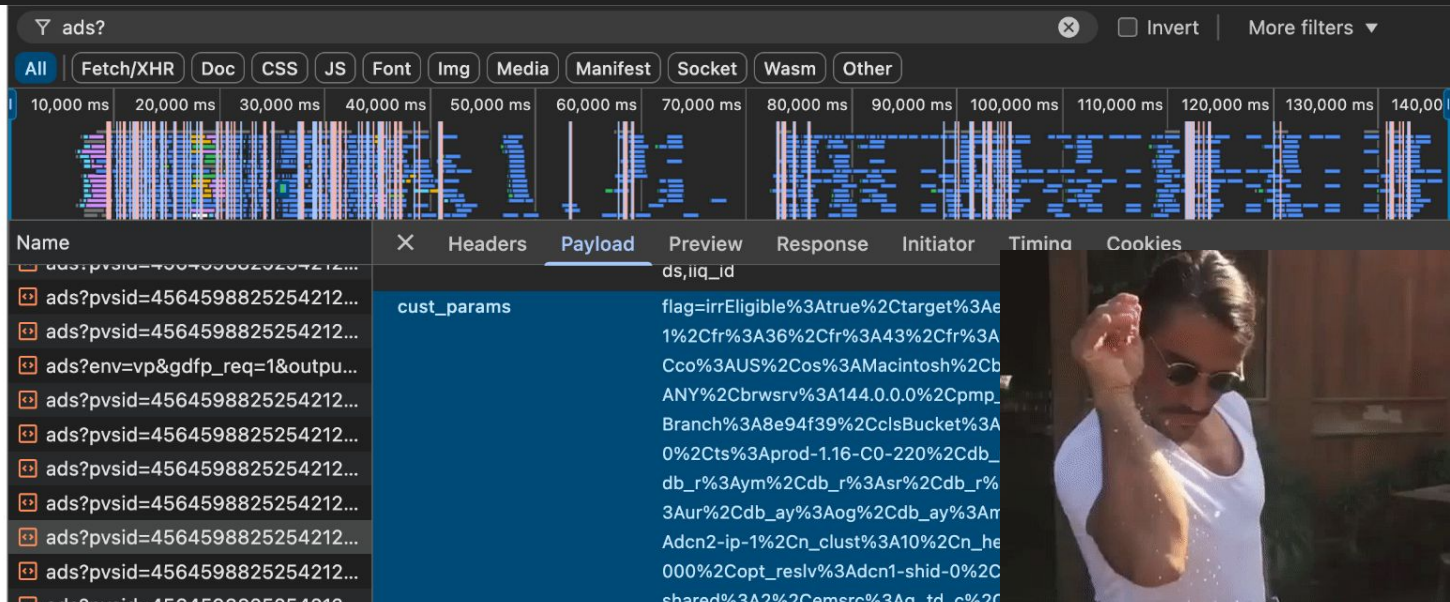


Custom Targeting a.k.a. “Key-Values”

- Query params on your ad request that **GAM can be configured** to consume
- This is how GAM knows to serve Prebid bids
- Can also be used to serve custom models – **load them prerender!**



Winter
Comfort Food



The End of the YouTube Video

Thanks for watching!

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If you have any questions **leave a
comment...**or just **ask now**

